Coherence of Resilience Measures for the Atlantic Meridional Overturning Circulation Across a Hierarchy of Models

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Abstract

The Atlantic Meridional Overturning Circulation (AMOC) is a critical component of the global climate system, and its potential collapse one of the most consequential climate tipping events. Quantifying AMOC resilience, that is, its capacity to maintain the active circulation state under external perturbations, is therefore of high scientific and societal importance. Resilience is a multifaceted concept, encompassing many notions of transient system response. Whether these distinct measures yield consistent, and therefore trustworthy, signals across models of different complexity has not been systematically examined. Here, we address this question by analysing four models that span a hierarchy in physical complexity: (i) a 3-box salinity model, (ii) a two-dimensional latitude-depth Boussinesq model of the overturning circulation, and the Earth system models of intermediate complexity (iii) CLIMBER-X and (iv) PlaSim-LSG. Across the hierarchy, we compute model-specific implementations of four complementary resilience concepts: local resilience, convergence time, basin volume, and minimal critical shock. Our results suggest that not all measures are coherent across the model hierarchy, as exhibited by incompatible

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trends under increasing CO_2 forcing in each model. This points to deficiencies in model compatibility and challenges in applying these concepts in real-world systems. Several measures indicate an unexpected increase in system resilience under climate change. These findings have direct implications for the detectability of an imminent AMOC collapse from observational or model-based early-warning signals.

Keywords: Atlantic Meridional Overturning Circulation, tipping points, resilience, model hierarchy

1 Introduction

The Atlantic Meridional Overturning Circulation is one of the dominant features of the Earth's ocean, transporting large amounts of warm, saline surface waters northward across the Atlantic and returning cold, deep North Atlantic water southward at depth (Talley et al., 2003; Johns et al., 2023). This large-scale transport pattern facilitates the relatively mild climate of Western Europe and plays a central role in the global carbon and heat cycles (Ganachaud and Wunsch, 2000; Jackson et al., 2015; Schaumann and Alastrué de Asenjo, 2025). Palaeoclimate records document repeated rapid transitions during the Pleistocene, so-called Dansgaard–Oeschger events, that are believed to have been accompanied by transitions between an active (AMOC-on) and a weakened or collapsed (AMOC-off) state (Dansgaard et al., 1984; Clark et al., 2002; Menviel et al., 2020). Under continued anthropogenic forcing, rising sea surface temperatures, freshening of the North Atlantic from increased precipitation, and Greenland ice-sheet melt may push the AMOC towards a similar transition (Yang et al., 2016; Lohmann and Ditlevsen, 2021; Armstrong McKay et al., 2022; van Westen et al., 2024).

More concretely, in background climate configurations where at least two stable equilibria coexist (the on and off states), the system can be permanently tipped from one to the other by crossing a critical threshold leading to the annihilation of one of the formerly stable states (bifurcation-induced tipping) or by a sufficiently large state perturbation (shock-induced tipping) (Ashwin et al., 2012; Alkhayuon et al., 2019). Rate-induced tipping is another pathway, though it will not be relevant for this work. In the bifurcation-theory framework for AMOC tipping, the transition to the off state is often conceptualised as a saddle-node bifurcation (Stommel, 1961; van Westen et al., 2025). A shock-induced transition, on the other hand, can be understood as an exit of the system state from the basin of attraction associated with the AMOC on state. In both frameworks, the AMOC is investigated as an autonomous dynamical system that may be influenced by slow parameter changes or external perturbations, such as noise. This allows for the characterisation of the system in terms of several quantities, subsumed under the term "resilience" (Krakovská et al., 2024).

One important notion of resilience has been applied to the AMOC in the context of a bifurcation-induced transition. As the system approaches such a transition through a gradual change in the background climate, the stable on-state's attractive forces weaken, a phenomenon known as critical slowing down (CSD) (Scheffer et al., 2009). CSD manifests in observable statistics: near the bifurcation, the dominant eigenvalue of the linearised system approaches zero, leading to increases in variance and autocorrelation of system fluctuations (Held and Kleinen, 2004; Dakos et al., 2008). These generic signatures have motivated a large body of research on early-warning indicators of tipping (Lenton, 2011; Boers, 2021; Ditlevsen and Ditlevsen, 2023).

Yet CSD-based indicators capture only the system's reaction to small perturbations around the equilibrium, known as local resilience, quantifying how the on-state changes locally as the bifurcation is approached. A complementary perspective is provided by non-local resilience concepts, which characterise the size and geometry of the basin of attraction of the on-state and therefore also capture the system's capacity to recover from larger perturbations (Holling, 1973; Gunderson, 2000; Menck et al., 2013). A comprehensive assessment of the resilience of the AMOC on state should encompass many different notions. In this work, we consider four resilience measures with the following heuristic definitions and interpretations. The concrete implementation and calculations for each model will vary and are explained in Section 2.

1. **Local resilience**: capacity of a system to continuously absorb small perturbations to its equilibium on-state, such as noise and internal variability (Arnoldi et al., 2016).

2. **Convergence time**: duration trajectories take to return to the on-state equilibrium after a large perturbation (Neubert and Caswell, 1997).

3. **Basin volume**: the volume fraction of a reference region of initial model conditions that converges to the on-state (Menck et al., 2013).

4. **Minimal critical shock**: the smallest instantaneous perturbation that moves the system out of the basin of attraction of the on-state (Klinshov et al., 2015; Halekotte

and Feudel, 2020).

In natural systems, controlled perturbation experiments are rarely feasible, making direct estimation of non-local resilience quantities from observations difficult. This is especially true for the AMOC, and numerical models, therefore, play an essential role. However, climate models span an enormous range of complexity, from analytically tractable conceptual models to full Earth system models (ESMs) with millions of degrees of freedom. It is not obvious whether resilience measures computed in simple models translate to their more complex counterparts, which include many additional processes and spatial heterogeneity (Stouffer et al., 2006; Schneider et al., 2017). If measures disagree across the hierarchy, it could mean that parts of the hierarchy do not expose them well. The extrapolation of findings from simple models to more complex ones, or to the real-world system, is then harder to justify. Several studies have examined individual resilience concepts in the context of the AMOC at specific levels of model complexity (Rahmstorf et al., 2005; Cimatoribus et al., 2014; Boers, 2021; Ditlevsen and Ditlevsen, 2023; Jacques-Dumas et al., 2024), but a systematic, multi-measure comparison across a model hierarchy is lacking.

Our analysis combines four modelling studies that span many orders of magnitude of model complexity: (i) a conceptual 3-box salinity model (Wood et al., 2019; Alkhayuon et al., 2019), (ii) a latitude-depth Boussinesq partial differential equation model (Dijkstra and Molemaker, 1997; Soons et al., 2025; Esclapez et al., 2026), and the Earth system models of intermediate complexity (iii) CLIMBER-X (Willeit et al., 2022) and (iv) PlaSim-LSG (Börner et al., 2026). Section 2 describes each model and the concrete resilience measures computed for it. Section 3 reports the individual model's results and Section 4 synthesises them across the hierarchy. In Section 4, we discuss implications for research on the AMOC and on early-warning detection of potential tipping.

2 Methods

We compute a common set of resilience measures for each model while respecting the different experimental possibilities at different levels of the hierarchy. More concretely, we analyse trajectories from perturbed initial conditions. The reduced state space in which perturbations and convergence are investigated is defined by North Atlantic and Tropical Atlantic salinity values. Salinity is conserved by applying a compensating perturbation in the remaining parts of the world ocean. Figure 1 illustrates the North Atlantic and Tropical Atlantic perturbation and diagnostic regions. The model-specific implementation, including salt compensation, is explained in the respective subsections below. Table 1 gives an overview of the concrete meaning and estimation routine for each resilience measure across the models. Initial conditions in the 3-box, Boussinesq, and CLIMBER-X experiments are probed within rectangles bounded by $\pm 2\,\mathrm{psu}$ perturbations relative to their model-specific reference configurations. This is done by drawing a large number of uniform samples from this region (3-box model) or by evenly sampling a grid of initial conditions (Boussinesq model and CLIMBER-X). PlaSim-LSG is an exception here, since we analyse existing edge-track and equilibrium runs from Börner et al. (2026) using two salinity readouts in the North and Tropical Atlantic. These trajectories constitute a specific kind of perturbed initial-condition experiment, as they record the evolution from the unstable edge state separating the on- and off-states to the on-state.

We repeat the resilience analysis across a global-warming forcing coordinate, as specified for each model below. This allows us to compare directional trends of resilience, which are more meaningful across models than absolute values expressed in model-specific units. The resilience of the AMOC on-state is the primary focus throughout, though all analyses could also be carried out for other stable equilibria when present.

2.1 3-Box salinity model

We adopt the 5-box salinity model introduced by Wood et al. (2019) and further reduced to 3 boxes by Alkhayuon et al. (2019). The original model partitions the Atlantic–Indo-Pacific basin into five boxes: North Atlantic (N), Tropical Atlantic (T), Southern Ocean

Table 1: **Overview of resilience-measure implementations across the model hierarchy.**

	3-Box model	**Boussinesq**	**CLIMBER-X**	**PlaSim-LSG**
AMOC strength	Overturning flux q at the on-state fixed point (see SI).	Signed largest-magnitude streamfunction extremum at 26°N, positive for northern sinking.	Maximum of the Atlantic overturning streamfunction at 26°N; trailing 500-year mean of the equilibrium run.	Maximum of the Atlantic overturning streamfunction between 46° and 66°N; mean over the on-state equilibrium run.
Local resilience	$-\mathrm{Re}(\lambda_{\max})$: negative real part of the least stable eigenvalue of the Jacobian at the on-state fixed point, evaluated via forward-mode automatic differentiation (Khan and Barton, 2015).	$-\max_k \mathrm{Re}[\log(\mu_k)/\Delta t]$, from the multipliers μ_k of a 2×2 map fitted to eight isotropic salinity perturbations.	$-\max_k \mathrm{Re}[\log(\mu_k)/\Delta t]$, from the multipliers μ_k of a 2×2 map fitted to eight isotropic salinity perturbations.	Inverse of the long axis of the on-state fluctuation cloud ellipse.
Convergence time	Mean return time of on-state trajectories among 10 000 uniform random initial conditions in a ±2 psu box; convergence at $\varepsilon = 0.05$ psu proximity.	Mean final observed entry into the 0.05 psu salinity-plane ball, conditional on on-state return, on a 20×20 grid over ±2 psu.	Mean return time of completed trajectories on a 9×9 grid over ±2 psu; convergence defined by permanent entry into an $\varepsilon = 0.05$ psu ball.	Mean transit time from the last edge-state 4σ ellipse visit to the first on-state 4σ ellipse entry.
Basin volume	Fraction of 10 000 random initial conditions in a ±2 psu box centred on the on state that converge to the on state.	Fraction of all 400 initial conditions on the 20×20 perturbation grid converging to the on state.	Fraction of the 9×9 grid initial conditions over ±2 psu that converge back to the reference equilibrium (permanent entry into an $\varepsilon = 0.05$ psu ball).	Not estimated because the available edge tracks do not thoroughly sample a reference region of state space.
Minimal critical shock	Minimum salinity-plane distance from the on state to a sampled initial condition not converging to the on state.	Minimum salinity-plane distance from the on state to a sampled initial condition not converging to the on state.	Minimum salinity-plane distance from the on-state to a sampled initial condition not converging to the on state.	Gap between the on-state and edge-state 4σ Gaussian ellipses along the line joining their centres in the deep-box salinity plane.

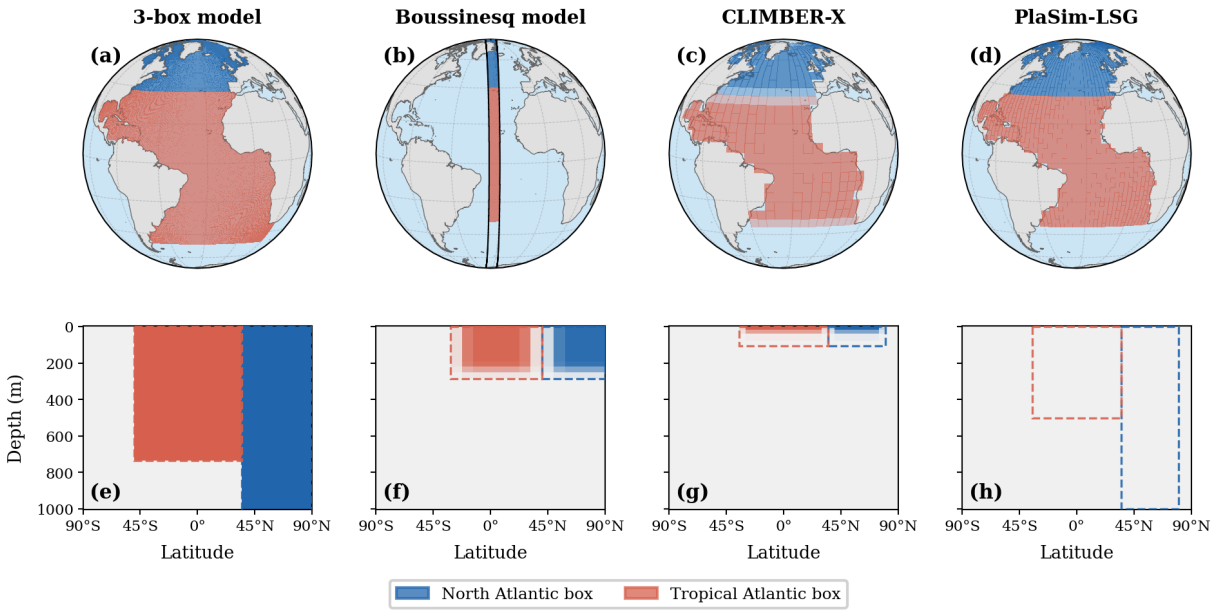


Figure 1: **Perturbation and read-out geometry across the model hierarchy.** Blue denotes the North Atlantic region and red the Tropical Atlantic region. Globe panels show their geographic footprints; latitude–depth sections show the perturbation fields (shading) and salinity-readout boundaries (dashed outlines). The solid black outline in the Boussinesq globe marks the full pole-to-pole model extent; its width is schematic. Boussinesq depths use a nominal basin depth of 1000 m, rather than a physical calibration. In PlaSim-LSG, the boxes are used only as salinity readouts. Compensation regions are omitted; the salt-conservation procedures are described in the text. The 3-box geometry is approximately reproduced from the model conception in Wood et al. (2019).

(S), Indo-Pacific (IP), and a Deep box (B). In the implemented 3-box reduction, the North and Tropical Atlantic salinity anomalies, S_N and S_T, are the two dynamical state variables. The slow variables of Southern Ocean and Deep-box salinities are prescribed as constants by the parameter vector, while Indo-Pacific salinity is derived at each time from salinity conservation. The exact model equations and parameters are given in the Supplementary Information.

Wood et al. (2019) calibrated two distinct parameter sets for the model: one for a pre-industrial climate ($p_{1\times}$) and one for a doubled-CO_2 climate ($p_{2\times}$) in the FAMOUS Earth System model. We construct a continuum of models through the full parameter vector as $p(r) = p_{1\times} + r(p_{2\times} - p_{1\times})$ and label it by $C(r) = 280\,\text{ppm} + 280\,\text{ppm}\,r$. Thus $r = 0$ and $r = 1$ correspond to 280 and 560 ppm, respectively, while $r > 1$ denotes linear extrapolation beyond the doubled-CO_2 parameter vector. The nominal spacing is $\Delta r = 0.05$, or 14 ppm, and we investigate the system from 280 through 896 ppm, with

one additional point at 903 ppm ($r = 2.225$). We find that the on-state vanishes in a bifurcation when extrapolating the linear parameter ramp further, between 903 and 910 ppm.

All numerical experiments and resilience calculations are performed using the Julia package Attractors.jl (Datseris et al., 2023) within the DynamicalSystems.jl software ecosystem. This computational framework enables efficient integration of large initial-condition ensembles and simultaneous tracking of multiple resilience measures as a parameter changes (Morr et al., 2026). At each continuation step, we probe the two-dimensional state space with 10 000 initial conditions sampled uniformly from a ± 2 psu rectangle centred on the on-state attractor. From these, we derive basin volume as the fraction of trajectories converging to the on state, the minimal critical shock as the minimum distance of the on state to an initial condition outside the basin, and convergence time as the mean time to reach a 0.05 psu neighbourhood. Local resilience is computed from the numerically derived Jacobian of the model's right-hand side at the on-state position. The resilience is measured as $-\mathrm{Re}(\lambda_{\max})$ where $\lambda_{\max}$ is the eigenvalue with the largest real part. In Figure 2, we map out the local basin geometry on a regular grid and show several example trajectories in terms of AMOC strength and phase-space evolution.

2.2 Boussinesq latitude-depth model

The second model in our hierarchy is a latitude-depth Boussinesq model of the meridional overturning circulation. The model evolves temperature, salinity and vorticity on a 21×41 grid of discretised latitude and depth points, with streamfunction diagnosed from vorticity. The surface layer is subject to thermal restoring and a prescribed salinity flux, with cooling and salt removal at high latitudes and warming and salt addition in the tropics. This model is much less expensive than global circulation models but already contains spatially structured salinity fields and overturning, making it a useful intermediate step between the conceptual box model and the three-dimensional climate models.

We introduce a forcing parameter γ that reduces the equator-to-pole thermal contrast by a factor $1 - \gamma$ and increases the symmetric salinity-flux amplitude by $1 + \gamma$. We use

$C = 280\,\text{ppm}\ 2^{\gamma/0.07}$ as an illustrative CO_2 coordinate. The coefficient is motivated by the reduction in the tropical-to-polar atmospheric temperature contrast in CMIP6 simulations (Hahn et al., 2021); the derivation and its assumptions are given in the SI. The equal fractional change in freshwater forcing remains a prescribed model coupling. We choose 20 forcing values between $\gamma = 0$ and 0.11, with finer spacing near the upper end. The on-state equilibrium is found up until $\gamma = 0.1085$, or approximately 820 ppm. Beyond this value, we find no on-state equilibrium, potentially due to a bifurcation in the high-dimensional dynamical system. We find two alternative equilibria over a large range of forcing values: a reversed-circulation equilibrium and a symmetric two-cell equilibrium with polar upwelling and equatorial sinking. The governing equations, equilibrium descriptions, and the numerical implementation are given in the Supplementary Information.

Initial-condition perturbations are applied to grid-aligned salinity boxes, using a fixed affine conversion from non-dimensional salinity to psu (we again refer to the SI for details). The North Atlantic box covers 40.5°–90°N and the Tropical Atlantic box 31.5°S–40.5°N. Both span the upper 28.449% of the model depth. Cosine-tapered anomalies change the two box means by $(\Delta S_N, \Delta S_T)$ on a 20×20 grid spanning ± 2 psu in each coordinate. A uniform compensating anomaly outside both boxes, including the deeper water, conserves total salt. Each of the 400 perturbations is applied to the on-state salinity field under the same respective forcing parameter value.

A trajectory is counted as returning to the on-state when its final two-box salinity lies within a Euclidean distance of 0.05 psu from the on-state equilibrium. For all 6800 grid trajectories, this classification agrees with an independent full-field convergence check (see SI). The mean convergence time is the mean final observed entry into that salinity neighbourhood, conditional on on-state return. Entries are resolved at intervals of 25 model years. Temporary passages through the neighbourhood do not establish convergence.

For local resilience, we run eight additional salinity perturbations of radius 0.01 psu, evenly spaced in angle around each on-state, together with an unperturbed control. These use the same perturbation geometry, salt compensation and complete equilibrium initial fields as the large grid experiment. Following the CLIMBER-X protocol, each run lasts

500 model years and is sampled annually. We centre the two-box salinities on the control mean over years 10–150 and fit a 2×2 map $x_{t+1} = Mx_t$ by least squares in that window. Its multipliers μ_k give the local resilience $r = -\max_k \mathrm{Re}[\log(\mu_k)/(1\,\mathrm{yr})]$. The smaller radius avoids the basin escapes observed for 0.1 psu near the upper forcing limit. This is an effective recovery rate in the two-box projection and depends on the fitting window; the full-state tangent spectrum is retained separately to verify equilibrium stability.

The AMOC strength is defined as the signed largest-magnitude extremum of $-\psi$ over depth at 26°N, so that northern sinking is positive and reversed circulation is negative. We apply a fixed transport normalization that assigns 17 Sv to the unforced on-state at 26°N, motivated by the approximately 17 Sv observed by RAPID near 26.5°N (Johns et al., 2023). This factor is held fixed across forcing values and circulation states; it is an amplitude normalization rather than a dimensional calibration of the idealized basin. We also use a fixed salinity conversion and a scale of 100 model years per non-dimensional time unit. Figure 3 shows four trajectories at $\gamma = 0$ and 0.09, labelled by 280 and 683 ppm.

2.3 CLIMBER-X

CLIMBER-X (Willeit et al., 2022) is a fast Earth system model of intermediate complexity (EMIC) that couples a 3-D frictional-geostrophic ocean model, a 2.5-D statistical-dynamical atmosphere, a dynamic-thermodynamic sea-ice model, and land-surface components. The ocean component resolves the full three-dimensional salinity and temperature fields, including the main overturning circulation cells, at a resolution of approximately $5° \times 5°$ with 23 vertical levels. CLIMBER-X is computationally efficient enough to permit multi-millennial integrations, making it well-suited for exploring the state space through moderately sized perturbation ensembles.

We first equilibrate CLIMBER-X at 15 prescribed atmospheric CO_2 concentrations between 285 and 600 ppm. Each equilibrium run is integrated for up to 20 000 model years from the default pre-industrial restart. After equilibration, the trailing 500 years of the annual ocean time series are used to store reference values for AMOC strength at 26°N and for the salinity diagnostics in the geographical boxes illustrated in Fig. 1.

The North Atlantic box comprises Atlantic-basin grid cells from 35°–80°N and the Tropical Atlantic box comprises Atlantic-basin cells from 35°S–35°N. Both boxes cover the upper 105 m, corresponding to the top four ocean layers. The North and Tropical box perturbations use a two-grid-cell latitudinal taper and a two-layer vertical cosine taper. The tapered fields are normalised so that the volume-weighted mean anomaly over the hard target box remains exactly equal to the prescribed salinity perturbation. The salt added to the two Atlantic boxes is compensated uniformly over the entire world ocean within the same upper 105 m, so that the global ocean salt content in the restart is conserved.

For our experiments, we apply instantaneous salinity perturbations to these boxes directly to the ocean restart as salinity anomalies in psu, creating an ensemble of modified initial conditions. The North Atlantic and Tropical Atlantic perturbations each take the values $-2, -1.5, \ldots, 2$ psu, giving a 9×9 grid of 81 experiments per CO_2 level. Each perturbed initial condition is integrated forward for 10 000 model years. This model is known to exhibit an AMOC off-state under high CO_2 concentrations when large constant freshwater forcing in the North Atlantic is applied. In the present setup of pure initial salinity perturbations, the off-state is not reached. Instead, at higher CO_2 a second, distinct attractor with a stronger AMOC emerges. This occurs when sea ice in the Northern Hemisphere retreats and the deep-water convection driving the AMOC expands into areas farther north (Willeit and Ganopolski, 2024). We classify each grid trajectory by its final state: a run is assigned to the on-state if its final salinities lie within an $\varepsilon = 0.05$ psu ball of the unperturbed reference state.

A further CO_2-dependence concerns the long equilibration run itself: the state on which it settles, and which serves as the unperturbed reference, changes branch with forcing. Up to 330 ppm it lies on the reference ("modern") branch, whereas from 345 ppm onward it lies on the stronger-AMOC ("strong") branch. The resilience measures obtained from the perturbation experiments are therefore associated with the modern state at 285–330 ppm and with the strong state at 345–600 ppm. The basin volume reported for CLIMBER-X (Table 1) is the fraction of the 81 grid trajectories that return to this

reference equilibrium. It equals one wherever all perturbations recover and falls below one only near the modern-branch transition (at 315 and 330 ppm), where a fraction of perturbations settle on the strong state instead. From the same grid, we obtain the minimal critical shock as the smallest salinity-plane distance from the reference equilibrium to a perturbed initial condition that does not return to its basin. It is defined only where at least one perturbation escapes, i.e. at 315 and 330 ppm, since at all other CO_2 levels the entire ± 2 psu grid recovers. Figure 4 shows four example trajectories starting from perturbed salinity initial conditions in the specified boxes, similar to the two figures above. For both CO_2 values, the equilibrium states are the modern state.

To estimate local linear stability, we run a second, smaller ensemble of 9 experiments per CO_2 level. Eight runs apply isotropic salinity perturbations of magnitude $\varepsilon = 0.1$ psu again arranged on a circle in the two-dimensional (NA, Tropical) shallow-box salinity space, plus one unperturbed reference run. Each run is integrated for 500 model years. An equilibrium reference is obtained from the mean of the unperturbed run over years 10–150. A 2×2 discrete linear map is fitted by least squares to annual trajectory pairs (x_t, x_{t+1}) in the same fitting window, whose multipliers μ_k yield the local resilience $-\max_k \mathrm{Re}[\log(\mu_k)/(1\,\mathrm{yr})]$.

2.4 PlaSim-LSG

The last component of the hierarchy is the coupled climate model of intermediate complexity PlaSim-LSG, comprising a spectral moist primitive-equation atmosphere (T21, $\approx 5.6^\circ \times 5.6^\circ$, 10 vertical levels) and the Large Scale Geostrophic (LSG) ocean (Maier-Reimer et al., 1993), a full 3-D ocean circulation model at $3.5^\circ \times 3.5^\circ$ resolution with 22 vertical levels, coupled to a thermodynamic sea-ice scheme and a dynamic hydrological cycle. With approximately 10^5 degrees of freedom and a throughput of roughly 700 model years per CPU day, PlaSim-LSG is considerably more complex than CLIMBER-X. In particular, it resolves atmospheric variability more faithfully and thereby yields chaotic ocean dynamics.

At 285 ppm (pre-industrial) and 360 ppm (contemporary) CO_2, PlaSim-LSG exhibits

a bistable AMOC with an on-state of $\approx 15\,\mathrm{Sv}$, measured as the streamfunction maximum in the latitude band 46°–66°N, and an off-state (deep convection in Subpolar Gyre ceases) of $\approx 1-2\,\mathrm{Sv}$, both confirmed by long unforced simulations. Börner et al. (2026) applied an edge-tracking algorithm (Lucarini and Bódai, 2017) to locate the edge state (Melancholia state) of the AMOC in PlaSim-LSG at 285 ppm and 360 ppm. This is a unstable equilibrium on the complex basin boundary between the on and off states. The algorithm iteratively bisects pairs of initial conditions converging to opposite attractors and tracks the resulting pseudotrajectory along the basin boundary until it converges to the edge state. The edge state is a chaotic saddle characterised by persistent centennial AMOC oscillations ($\approx 120\,\mathrm{yr}$ period, amplitude 3–10 Sv), driven by atmosphere–ice–ocean interactions in the North Atlantic. A complete description of the model configuration, its AMOC bistability, and the edge-tracking procedure is given in Börner et al. (2026), to which we refer the reader for full methodological details.

From the final iterations of edge tracking at each CO_2 level, we extract pairs of edge tracks: one converging to the on-state and one converging to the off-state, both initialised from near-boundary conditions that closely shadow the edge state before diverging. These trajectories provide a natural set of initial conditions from which resilience-relevant transient behaviour can be quantified: they start close to the basin boundary and thus represent challenging recovery paths for the on-state. In total, we analyse 37 paired edge-track files at 285 ppm and 38 at 360 ppm. Dedicated long equilibrium simulations of the AMOC-on, AMOC-off, and edge states at each CO_2 level provide the attractor clouds against which these transients are evaluated.

We reduce the high-dimensional PlaSim-LSG salinity output to the salinity of two geographical boxes. The North Atlantic readout is the mean salinity from 35°–80°N and 0–1000 m depth, and the Tropical Atlantic readout covers 35°S–35°N and 0–500 m depth. Dedicated on-, off-, and edge-state equilibria trajectories determine the means associated with these states in this two-dimensional salinity plane. The covariance matrices of the scattered trajectory points are computed to obtain an approximate Gaussian distribution of data for each state. Convergence time is measured from the last visit to the edge-state

4σ Gaussian ellipse to the first entry into the target-attractor ellipse, and the critical-shock proxy is the gap between the edge-state and on-state ellipses, measured along the line joining their centres. Local resilience is quantified as the inverse width of the on-state fluctuation cloud, $[2000\sqrt{\mu_{\max}(C)}]^{-1}$, where $\mu_{\max}(C)$ is the largest eigenvalue of the on-state equilibrium covariance C. A broader cloud corresponds to a smaller variability-based resilience proxy. If the centred salinity readouts admit a stable linear stochastic approximation $\mathrm{d}X = AX\,\mathrm{d}t + B\,\mathrm{d}W_t$, their stationary covariance satisfies the Lyapunov equation

$$AC + CA^\top + BB^\top = 0 \tag{1}$$

(Arnoldi et al., 2016). There is no general one-to-one relation between the leading eigenvalues of C and A. Under the additional assumptions of normal dynamics ($AA^\top = A^\top A$) and isotropic noise ($BB^\top = \sigma^2 I$),

$$\mu_{\max}(C) = \frac{\sigma^2}{-2\,\mathrm{Re}(\lambda_{\max}(A))}, \tag{2}$$

where $\lambda_{\max}(A)$ has the largest real part. At fixed noise intensity, the inverse-width proxy therefore increases with the square root of the linear decay rate. Changes in effective noise or departures from these assumptions can alter this relationship, so the proxy does not directly estimate a dynamical eigenvalue.

3 Results

The trajectories from the described perturbation experiments are synthesised to give resilience estimates according to all measures listed in Table 1. The evolution of the resilience measures is shown in Figure 6, which forms the basis for the subsequent comparison and differentiation of the results. The models differ substantially in the dynamical characteristics of the AMOC, including its stationarity, multistability, and bifurcations. This, in itself, complicates the interpretation of our results on AMOC resilience within a common framework.

The models do not agree on a consistent impact of atmospheric CO_2 on the equilibrium AMOC strength. The 3-box model exhibits AMOC weakening with increasing CO_2, which is consistent with behaviour exhibited by the FAMOUS Earth System Model (Bouttes et al., 2015), the calibration target for the 3-box model (Alkhayuon et al., 2019). The vanishing of the AMOC on-state in a bifurcation at a threshold CO_2 concentration has, on the other hand, not been observed in FAMOUS. The Boussinesq on-state also weakens along its imposed forcing path, from 17.0 to about 15.1 Sv on the fixed reporting scale, an 11.2% decrease. No stable on-state was verified beyond $\gamma = 0.1085$; this numerical limit does not by itself establish a bifurcation. In CLIMBER-X and PlaSim-LSG, AMOC strength instead increases as the CO_2 concentration rises. For these two models, it has been conjectured that this is due to a changing net surface freshwater flux, particularly in the Tropical Atlantic (Willeit and Ganopolski, 2024). More evaporation implies more saline water reaching the North Atlantic. In the Boussinesq model, the imposed forcing amplifies the entire symmetric salinity-flux profile while reducing the thermal contrast, so it does not isolate this tropical freshwater mechanism. PlaSim-LSG exhibits a vanishing AMOC on-state at a CO_2 forcing of about 460ppm, while CLIMBER-X does not lose its AMOC on-state.

The existence and characteristics of an AMOC off-state are also not consistent across the model hierarchy. The 3-box and Boussinesq models feature stable states with reversed AMOC flow. That is, the salinity gradient between the North and Tropical Atlantic reverses in both models, and in the Boussinesq model, the actual fluid overturning changes direction. In the Boussinesq model, another neutral equilibrium with polar upwelling and equatorial sinking also attracts an increasing share of perturbations as forcing rises. It is unclear whether this is physically meaningful behaviour, i.e. whether any real-world AMOC collapse would lead to a strong reversed flow. However, as we regard alternative stable states primarily as differentiations from the contemporary AMOC state, their exact characteristics are of little concern. The PlaSim-LSG model exhibits an AMOC off-state characterised by a breakdown of deep-water convection in the North Atlantic. The CLIMBER-X model exhibits a weak AMOC state only at CO_2 concentrations lower than

preindustrial conditions. This is due to more extensive North Atlantic sea ice blocking deep-water convection in the Labrador and Irminger Seas. The only relevant alternative state for our experiments is a stronger AMOC state at large CO_2 concentrations, reached when diminished Northern sea ice allows for additional deep-water convection in the Kara Sea and Nansen Basin (Willeit and Ganopolski, 2024). It is important to consider these differences, especially when interpreting the results of basin volume and minimal critical shock. The two measures should only be interpreted as resilience against AMOC collapse when the alternative state is indeed associated with an AMOC weakening.

Local resilience of the on-state describes the restoring force that returns small perturbations back to the equilibrium. For a stable linear system, it describes the slowest asymptotic exponential decay rate; finite-time recovery can also depend on transient amplification and the observed variables. Critical slowing-down theory predicts a decrease in local resilience as one approaches a bifurcation tipping point. This is indeed what we see for the 3-box model. Local resilience of the CLIMBER-X AMOC also seems to eventually decrease with increasing CO_2. In the Boussinesq model, the projected estimate first increases from $0.0121\,\mathrm{yr}^{-1}$ at $\gamma = 0$ to $0.0143\,\mathrm{yr}^{-1}$ at $\gamma = 0.05$, then decreases to $0.00259\,\mathrm{yr}^{-1}$ at the last verified point, $\gamma = 0.1085$. Its magnitude is sensitive to the fitting window, so it characterises recovery over the specified interval rather than an asymptotic full-state decay rate. For the CLIMBER-X model, the decrease is associated with large fluctuations and therefore less certain. In the PlaSim-LSG model, the on-state fluctuation cloud slightly decreases in size, yielding a larger covariance-based resilience proxy; this alone does not establish a change in the dynamical decay rate.

The convergence time aggregates generally increase with CO_2 forcing in the 3-box, CLIMBER-X and PlaSim-LSG models. Longer recovery can be interpreted as a loss of resilience, as the AMOC is outside its usual operating regime for longer. The Boussinesq model follows a different pattern: the mean projected return time falls overall from about 277 to 81 model years across the verified forcing range, with intermediate increases. These means condition on a shrinking set of returning perturbations. The 13 perturbations that return at every forcing value show a smaller overall decrease, from about 102 to 81 model

years. Recovery in the complete model state takes longer and increases near the last verified on-state (see SI), illustrating that projected return times can conceal slow recovery in other variables. One model-specific behaviour is worth emphasising in this context. The transient trajectories of some perturbations in the CLIMBER-X model at 315ppm show long occupancy of the strong AMOC state before eventually converging to the modern state (see SI Figure S2). It is possible that all perturbations marked as converging to the strong state in the 10,000 years of trajectory integration would eventually return to the modern state. Under the 330ppm configuration, these long transients remain in the strong state basin for the entire 10ky, reducing the estimated basin volume of the modern state and the mean convergence time. Whether this, too, constitutes spurious metastability cannot be determined with these experiments. This highlights the limitations in the numerical computation of resilience measures and the need to accurately define a finite-time framework for their interpretation. Throughout this study, convergence is never determined analytically but obtained via a finite-time cutoff evaluation against a neighbourhood convergence criterion (Morr et al., 2026).

A less intuitive result emerges from the basin volume computations. The share of initial perturbations in the region spanned by ± 2 psu in Northern and Tropical salinity that converge to the on-state rises in the 3-box model when CO_2 increases. This is surprising insofar as it indicates a strengthening of the AMOC's resilience in this interpretation. Only in the immediate proximity to the bifurcation threshold of the 3-box model do we see a sharp decrease in basin volume. In contrast, the sampled Boussinesq on-state basin contracts from 82.5% to 3.25% across the verified forcing range, as an increasing share of perturbations reaches the neutral circulation state. The basin structure of the CLIMBER-X model is almost trivial. For most considered CO_2 concentrations, all probed perturbations converge to either the modern AMOC on-state (300ppm and below) or the emerging strong AMOC state (345ppm onwards). Only at 315 ppm and 330 ppm do perturbations of the modern state lead to a transient convergence to the strong state, splitting the basin.

Finally, the minimal critical shock increases in the 3-box and PlaSim-LSG models. In

the Boussinesq model, the smallest sampled escaping shock instead falls broadly from 1.34 to 0.15 psu. In CLIMBER-X, it is only defined at 315 and 330 ppm, where it decreases as the strong state takes over a growing part of the probed region. The increase in the 3-box and PlaSim-LSG models would mean an increase in resilience, as larger perturbations are needed to push the AMOC to divergence away from the on-state.

4 Discussion

The picture emerging from our multi-model, multi-measure comparison of AMOC resilience is ambiguous. It is confounded by challenges in conceptual compatibility and practical differences in dynamical characteristics. Some measures exhibit internal inconsistencies in resilience trends across models. Within a model, faster local recovery need not coincide with greater resistance to finite perturbations, as the Boussinesq results demonstrate. The notion of a real-world AMOC at risk of bifurcation tipping due to anthropogenic influences suggests a decrease in all resilience measures discussed near the bifurcation point (Morr et al., 2026). The simple bifurcating 3-box model does produce a decrease in basin volume and minimal critical shock, but only as it enters the direct proximity of the bifurcation, and the on-state basin contracts to a smaller size than the ± 2 psu salinity region.

The decline in projected local resilience at higher forcing in the Boussinesq model, and the less systematic decline in CLIMBER-X, are compatible with slower recovery under warming. The dependence on the fitting window and projection limits their interpretation as evidence of an approaching bifurcation. However, these quantities are obtained in equilibrium, and the real-world AMOC is likely still adapting to fast anthropogenic influences. The direct implications for the applicability of CSD-based early warning signals, therefore, remain limited. In addition, the PlaSim-LSG result shows a reduction in stationary variability, whose relationship to a dynamical decay rate also depends on the effective noise. The increase in non-local convergence time in several models suggests slower recovery from larger perturbations as well. The dependence of Boussinesq return times on the observed variables and returning subset shows why finite-amplitude recovery

cannot be inferred from a local decay rate alone.

Basin volume, as we have computed it here, provides insight into how the on-state provenance evolves under the chosen perturbation regime. In the 3-box model, the initial volume of the on-state basin is consumed entirely by the off-state basin as the CO_2-forcing increases. For the Boussinesq model, 317 of the 330 initially returning grid perturbations instead reach the neutral circulation state at the highest verified forcing. Only 13 of all 400 perturbations still return to the on-state. The contraction of the modern state basin in the CLIMBER-X model reveals novel insights into Earth System Model dynamics. To our knowledge, this is the first instance of a model of this size exhibiting bistability with respect to changes in the initial salinity configuration. Typically, modelling studies employ persistent hosing influences to permanently alter surface salinity and facilitate an AMOC collapse. The possibility that pulse perturbations could similarly lead to the convergence to an alternative stable state, albeit a stronger one, should inspire future research.

The minimal critical shock provides context for the results on basin volume, since the two are mechanistically related. In the 3-box model, it reveals that the increase in basin volume is at least in part mediated by an expansion of the basin boundary at the point where it is closest to the on-state. This point is a natural candidate for the so-called instanton path to pass through (Grafke and Vanden-Eijnden, 2019; Soons et al., 2025). This is the most likely path in which a random transition occurs when forced by small noise. It is also related to saddle points or edge states, which are unstable equilibrium points on the boundary (Lucarini and Bódai, 2017; Lohmann et al., 2025). The result of the PlaSim-LSG model is consistent with the interpretation of a distancing saddle point on the basin boundary of the 3-box model. This is because, in this case, we are measuring the distance between the on-state and the edge state, which was directly approximated using an edge-tracking algorithm (Börner et al., 2026). The opposing trends in minimal critical shock should be studied in a broader selection of models going forward, as it may reveal important information about the most likely transition path in an AMOC collapse.

Our comparative study of AMOC resilience does not offer concrete physical insight into AMOC dynamics. However, the discrepancies and consistent behaviours found across

measures and models inspire a more careful selection of these measures and models. In principle, the evolution of resilience measures could provide valuable insight into the mechanistic changes the AMOC undergoes under climate change. It is not a priori expected that simple models exhibit the same resilience behaviour as complex models, and none may represent the true AMOC dynamics well enough to allow generalisation of our findings. Our findings offer a first insight into whether and how resilience can be consistently defined and computed across models of varying complexity. Experiments with higher-fidelity models may help resolve some of the discrepancies that arose here. If the real-world AMOC is indeed approaching a bifurcation threshold, we expect insights from the simple 3-box model to be valuable, as many current climate models struggle to reproduce this threshold (Valdes, 2011; Liu et al., 2017; McCarthy and Caesar, 2023). On the other hand, results from simple models should be regarded with caution and treated as strong abstractions.

We hope to inspire future research on the AMOC and other natural systems under the framework of dynamical system resilience. Its system-agnostic nature enables effective comparison of system dynamics, while questions remain about its consistent adaptability across model hierarchies.

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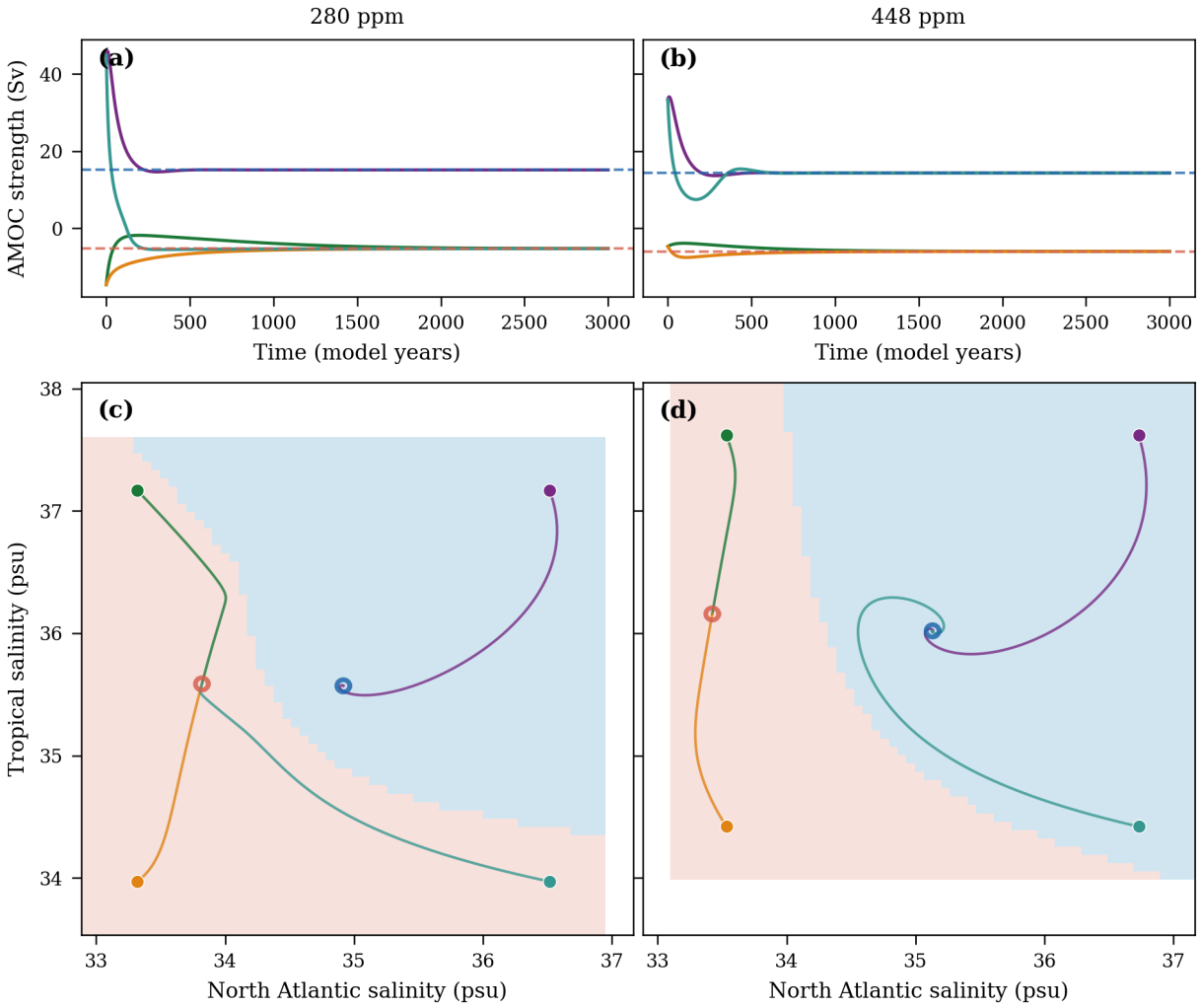


Figure 2: **Three-box model response to salinity perturbations.** Top panels (a, b) show the evolution of the AMOC strength for four representative initial conditions, with dashed horizontal lines marking the on- and off-state equilibria. Bottom panels (c, d) show the same trajectories in the two-dimensional salinity state space (the two state variables), overlaid on the sampled basins of attraction; circles represent 0.05 psu neighbourhoods around the on- and off-state fixed points (blue and orange, respectively). The left and right columns correspond to 280 and 448 ppm model specifications, respectively.

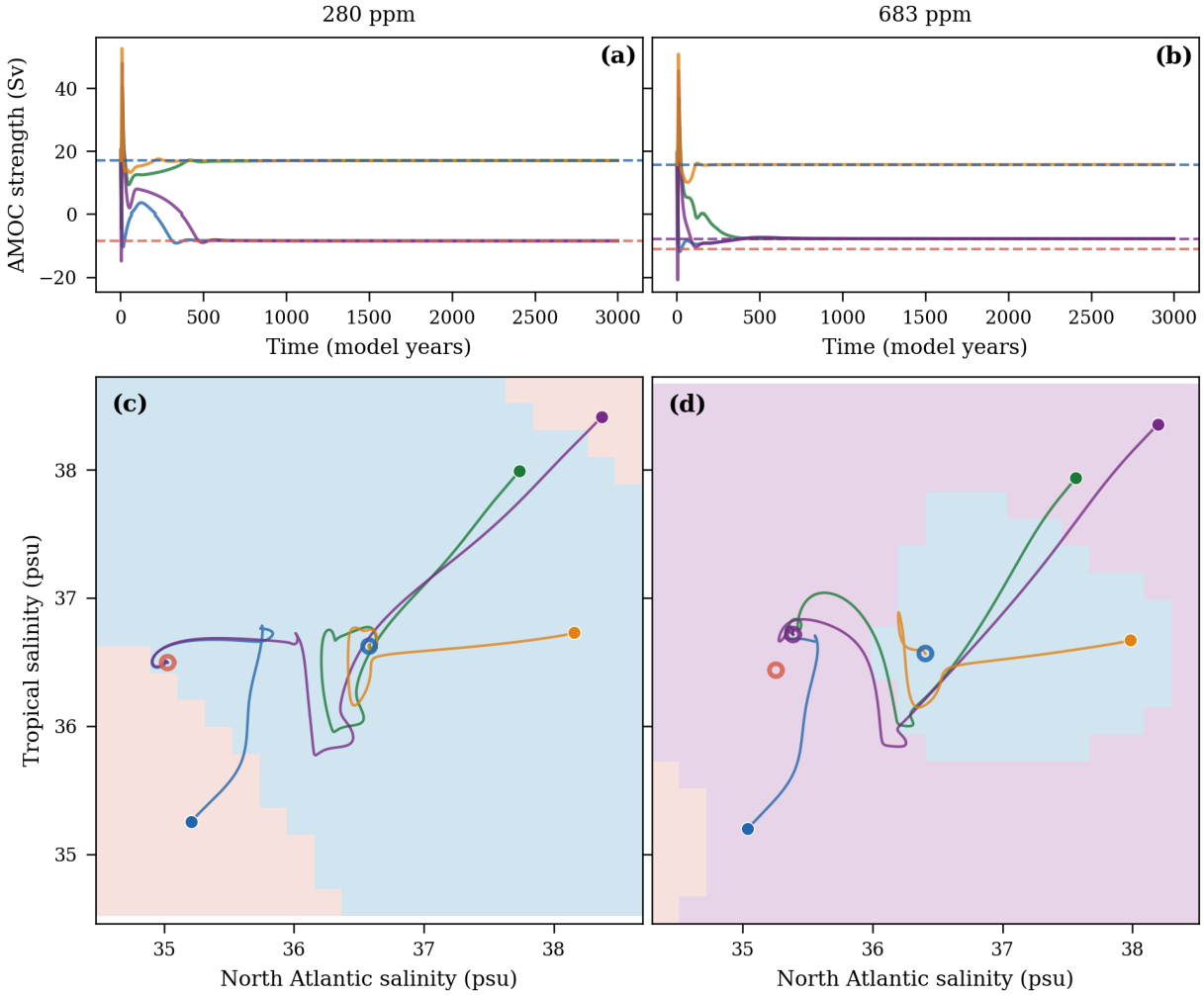


Figure 3: **Boussinesq model response to salinity perturbations.** Top panels (a, b) show the signed overturning strength at 26°N from the instant of four salinity perturbations to the complete on-state equilibrium. Bottom panels (c, d) show the same trajectories in North Atlantic and Tropical Atlantic box-mean salinity, overlaid on the sampled basins of attraction. Basin shading and circles identify the on state (blue), reversed state (red) and two-cell state (purple). Dashed lines mark equilibrium readouts; the reversed and two-cell states have negative values because their circulation at 26°N opposes northern sinking. Circles of radius 0.05 psu show the salinity-plane neighbourhoods used for return timing; endpoint classifications also agree with the independent full-field check. The columns correspond to $\gamma = 0$ and 0.09, labelled by 280 and 683 ppm. The same four anomalies are used in both columns; three reach the two-cell state in the right column and one returns to the on state.

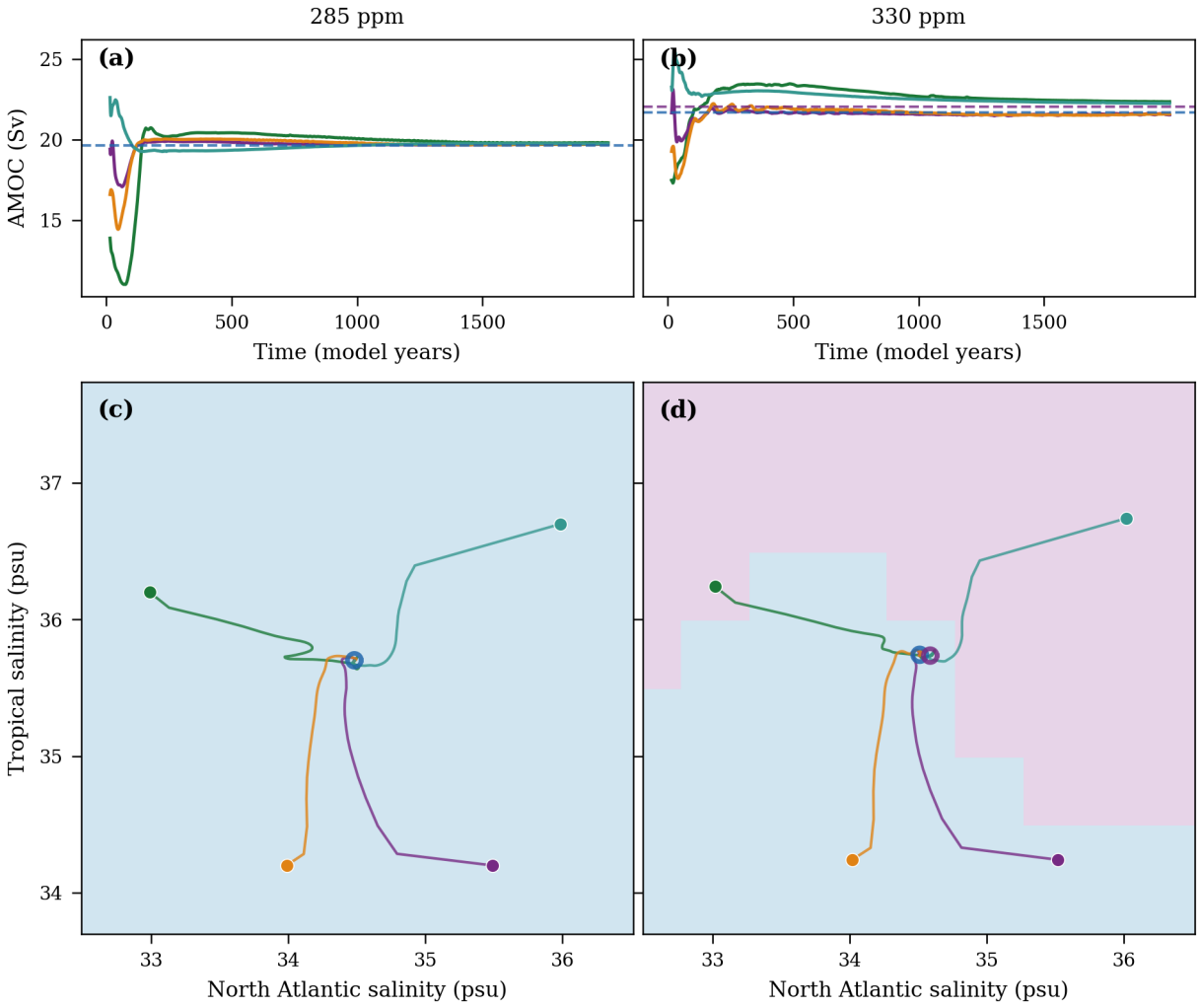


Figure 4: **CLIMBER-X model response to salinity perturbations.** Top panels (a, b) show the AMOC strength at 26°N along four example trajectories from perturbed salinity initial conditions. Bottom panels (c, d) show the same trajectories in the two-dimensional Atlantic-box salinity plane, with axes spanning exactly the ±2 psu perturbation window. The shaded background maps the basin of attraction of every initial condition on the full 9×9 grid: cells whose trajectory returns to the reference on-state (final salinity within $\varepsilon = 0.05$ psu) are blue, and cells that settle on the distinct stronger-AMOC ("strong") state are purple. Circles mark the $\varepsilon = 0.05$ psu neighbourhoods of the on-state (blue, at the unperturbed equilibrium) and of the strong state (purple, at the mean endpoint of the non-converged trajectories). The two columns show prescribed atmospheric CO_2 concentrations of 285 ppm and 330 ppm: at 285 ppm all 81 initial conditions return to the on-state, whereas at 330 ppm 46 of the 81 converge to the strong state, which lies only ≈0.08 psu from the on-state but exhibits an AMOC about 0.4 Sv stronger. Dashed horizontal lines in (a, b) mark the equilibrium AMOC strength of the on-state (blue) and, in (b), of the strong state (purple).

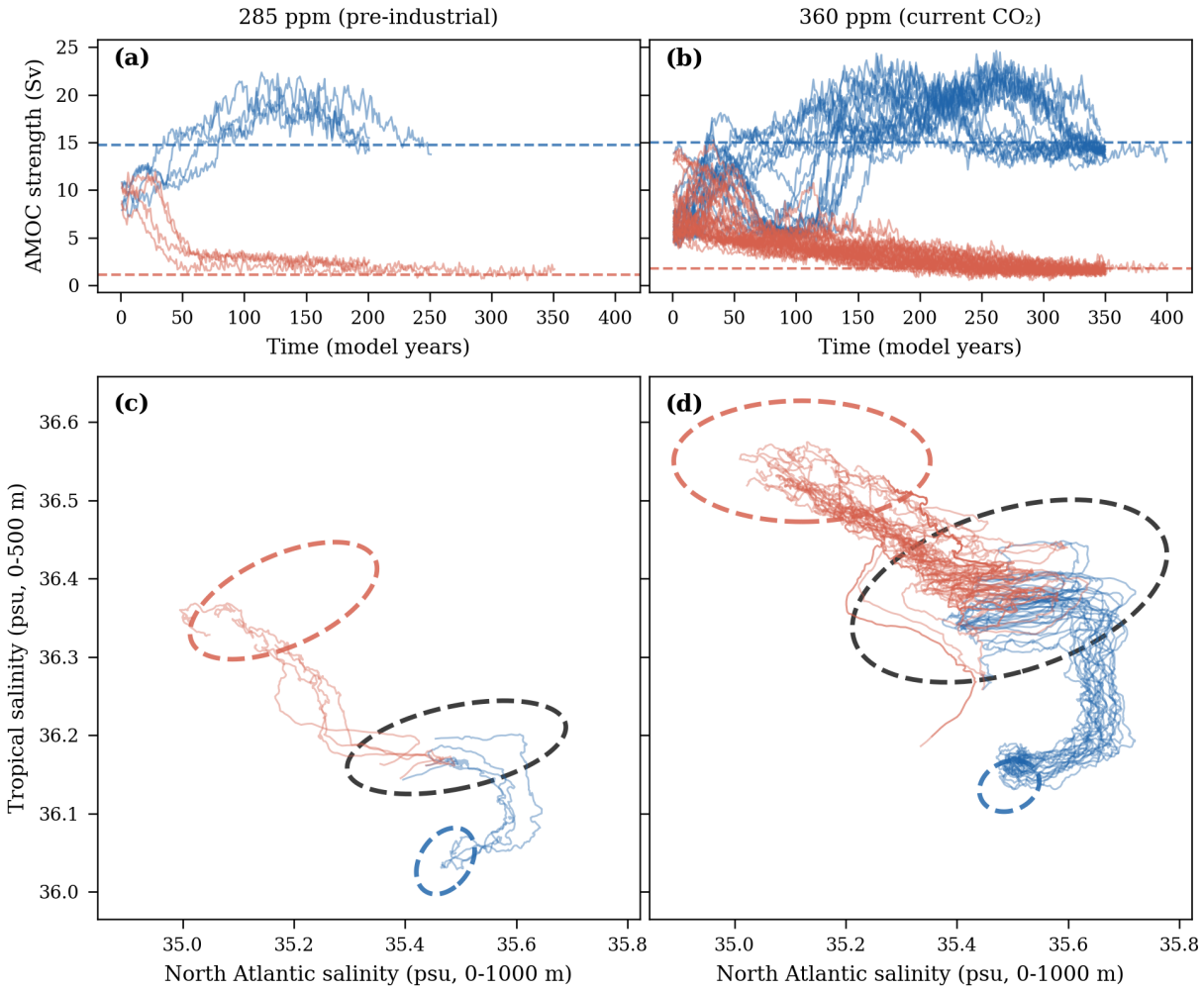


Figure 5: **PlaSim-LSG edge-state trajectories and covariance ellipses of attractors.** Top panels (a, b) show the AMOC strength readout along converged edge-track trajectories. Bottom panels (c, d) show the same transients in the North and Tropical Atlantic salinity coordinates. Ellipses denote the fitted 4σ Gaussian regions to equilibrium runs of the three states (on: blue, off: orange, edge: black), used to define convergence times and edge-to-attractor distances. Trajectories are coloured by the state they converge to. The two columns show the experiment results at prescribed atmospheric CO_2 concentrations of 285 ppm and 360 ppm, respectively.

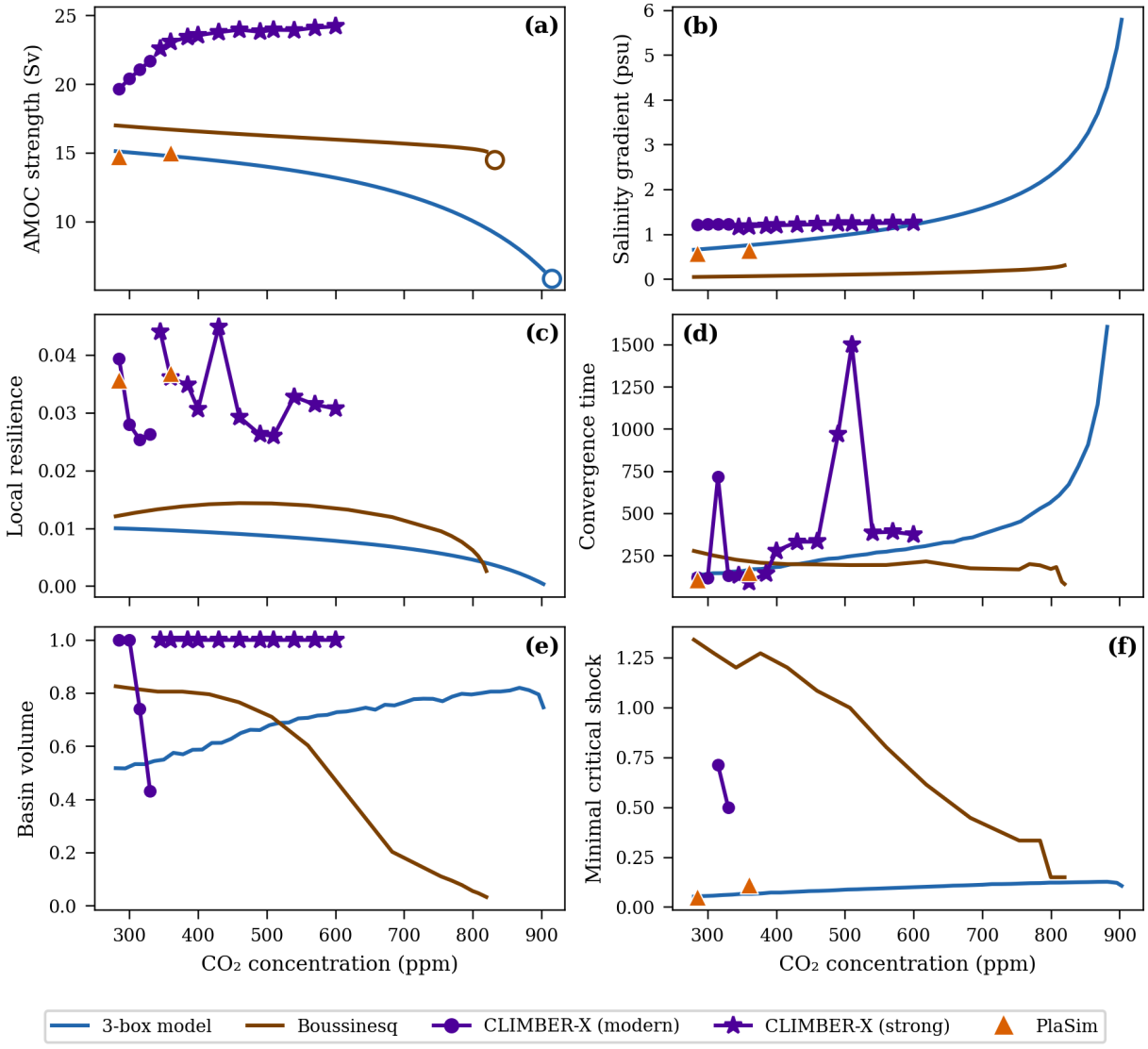


Figure 6: **Cross-model synthesis of AMOC resilience measures.** Equilibrium AMOC strength (a) and Tropical minus North Atlantic salinity gradient (b). Hollow circles in (a) mark the last plotted on-state points: the 3-box branch approaches a bifurcation, whereas the Boussinesq endpoint is the last verified stable equilibrium and does not establish a bifurcation. Boussinesq transport is normalized to 17 Sv at $\gamma = 0$ with one fixed factor across forcing values. Resilience quantifiers are plotted against the CO_2 coordinate for each model, where available (c–f). The model-specific definitions and implementation of each resilience quantifier are given in Table 1. These results provide the basis for assessing which resilience signals are coherent across the model hierarchy.